

Determining the Best Supplement for Pig Weight over time

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Study Context and Data Set

Study Context

- The data was collected from a total of 72 pigs from 24 litters which were assigned to combinations of vitamin E and copper treatments. Each pig was observed repeatedly, with weight recorded weekly over a 12-week period.

Objective

- To look at how two different supplement's(Copper and Vitamin E) impact pig growth over time.

Impact

- Optimizing Animal Growth and Agricultural productivity

Data Set Used

- Dietox dataset from R's doBy package
 - Dataset based on the following study
Lauridsen, Charlotte & Højsgaard, Søren & Sorensen, Martin. (1999). Influence of dietary rapeseed oil, vitamin E, and copper on the performance and the antioxidative and oxidative status of pigs. Journal of animal science. 77. 906-16. 10.2527/1999.774906x.

Data Set Details and Structure

Shape of Data

- 861 X 8

Rows

- Pigs at specific time point

Columns

- Weight - Weight in kg
- Feed - Cumulative feed intake in KG
- Time - Weeks in the experiment
- Pigs - ID of each Pig
- Evit - Vitamin E Dose
- Cu - Copper Dose
- Start - Starting Weight
- Litter - Specific Litter the pig was taken from

Data Set Details and Structure

Response Variable

- Weight(kg) of pig - measured weekly

Experimental Factors

- Vitamin E(3 levels) - 0, 100, 200 mg/kg
- Copper(3 levels) - 0, 35, 175 mg/kg
- Time(12 levels) - 1-12 Weeks

Experimental Unit

- Pig

Measurement Unit

- Pig at specific time point

Structure

- Design: Unbalanced Completely Randomized Design(CRD)
- Model Type: Mixed-effects model
- Factorial Treatment Structure (Evit x Cu x Time)
 - Repeated Measures on each pig

Pigs Per Treatment			
	Cu000	CU035	Cu175
Evit000	7	8	8
Evit100	8	8	8
Evit200	8	9	8

Data Set Details and Structure cont.

Fixed Effects

- Vitamin E, Copper and Time

Random Effects

- Litter
 - Shared genetic/environment
- Pig Nested within Litter
 - Individual variability between pigs

Key Features

- Observations are not independent
 - Repeated measures within pig
 - Correlation within litter

Unused Variables

- Start Weight
 - Individual variability accounted for at pig level
- Feed
 - Could obscure treatment comparisons

Theoretical Model

Linear Mixed-Effects Model

●
$$Y_{ijklm} = \mu + \alpha_i + \beta_j + \tau_k + (\alpha\beta)_{ij} + (\alpha\tau)_{ik} + (\beta\tau)_{jk} + (\alpha\beta\tau)_{ijk} + L_m + P_{l(m)} + \epsilon_{ijklm}$$

- y : (response) Weight of pigs
- μ : Overall mean weight of pigs
- α_i : Main effect of the i^{th} level of Vitamin E
- β_j : Main effect of the j^{th} level of Copper
- τ_k : Main effect of the k^{th} time point
- $(\alpha\beta)_{ij}$, $(\alpha\tau)_{ik}$, $(\beta\tau)_{jk}$: Two-way interactions
- $(\alpha\beta\tau)_{ijk}$: Three-way interaction
- L_m : Random effect of the m^{th} litter
- $P_{l(m)}$: Random effect of the l^{th} pig nested within the m^{th} litter
- i : 1,2,3 - 0, 100, 200 mg dl-alpha-tovopheryl acetat / kg feed
- j : 1, 2, 3 - 0, 35, 175 mg/kg feed
- k : 1, 2, 3, . . . , 12 - weeks
- l : 1, 2, 3, . . . , 21 - Litters
- m : 1, 2, 3, . . . , n_m - where n_m is the number of pigs in the m^{th} litter

Model Assumptions and Justification

Normally Distributed Errors

- $\varepsilon_{ijklm} \sim N(0, \sigma^2)$

Normally Distributed Litters

- $L_m \sim N(0, \sigma_L^2)$

Normally Distributed Pigs (Nested within Litters)

- $P_{l(m)} \sim N(0, \sigma_{P(L)}^2)$

Mutual Independence

- The random effects are all mutually independent of one another

Justification of Model Choice:

- Mixed-effects model used because pigs are measured repeatedly over time
- Random effects capture variability between pigs and litters

Note: This model has inappropriate assumptions and structure in the variance/covariance matrices due to us not covering repeated measures in time during the semester

Research Questions

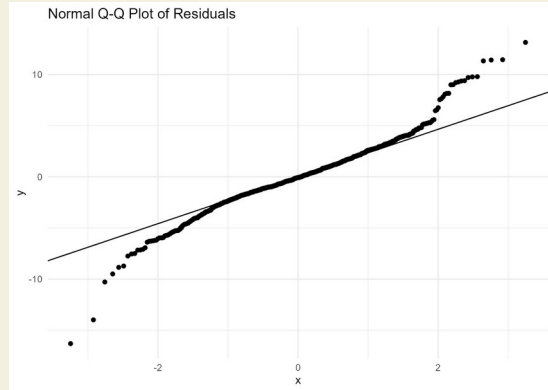
Why the data is interesting:

- Continuous growth tracked over time
- All interactions between treatments are present in the data
- Litter and pig information is present
- Has real-world applications

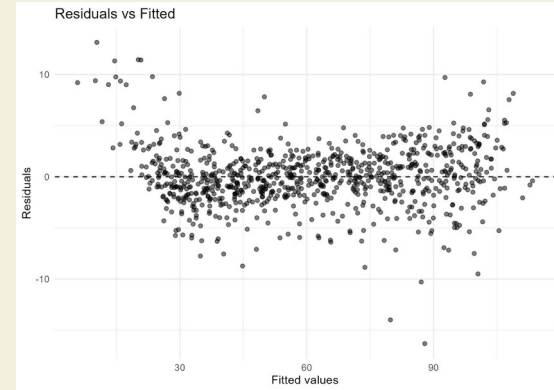
Research Questions:

- Does weight change over time?
- Do Vitamin E or Copper affect weight?
- Do Vitamin E and Copper interact?
- Are there treatment differences in average weight across groups?

Non-Transformed Data



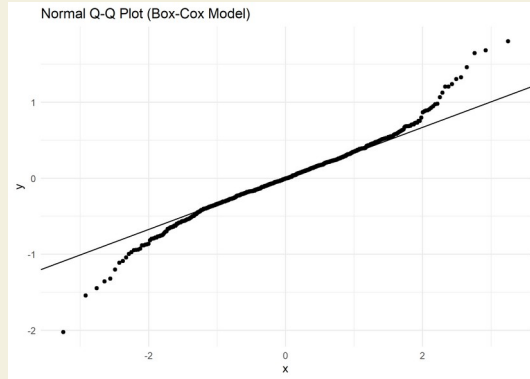
Shapiro-Wilks Test	
W	p_value
0.959867	1.29E-14



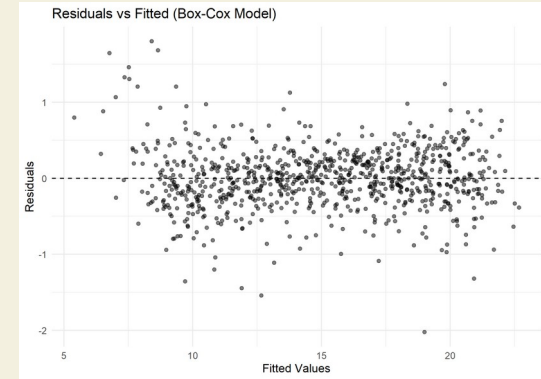
	Brown-Forsythe Test				
	Df	Sum Sq	Mean Sq	F value	Pr(>F)
group_var	107	513.0614	4.794966	0.999604	0.486479
Residuals	753	3612.04	4.796866	NA	NA

- Shapiro-Wilks Tests shows a significant p-value that would mean that we reject the hypothesis of normality, however, with a large sample size it is known that the test will be significant regardless of normality
- Looking at the Q-Q Plot and Residuals vs Fitted Values plot, assumed normality of data
- Brown-Forsythe Test was not found to be significant

Box Cox Transformation



Shapiro Wilks(Box-Cox)	
W	p_value
0.973196	1.75E-11



Brown Forsythe Box Cox					
	Df	Sum Sq	Mean Sq	F value	Pr(>F)
group_var	107	8.316848	0.077728	0.82613	0.892251
Residuals	753	70.84706	0.094086	NA	NA

- Completed a Box-Cox transformation in attempt induce more normality with $\lambda = 0.54$
- Marginal Improvements
- Same results
- Used original data for analysis for increased interpretability and less complexity

Evit X Cu Summary Statistics

Evit	Cu	n_obs	n_pigs	mean_weight	sd_weight	mean_start
Evit000	Cu000	84	7	61.24760	24.85101	25.55714
Evit000	Cu035	96	8	59.34373	25.02413	24.20000
Evit000	Cu175	96	8	61.23332	25.97867	25.15000
Evit100	Cu000	95	8	61.43577	24.99730	25.88631
Evit100	Cu035	96	8	61.29998	25.52994	27.31250
Evit100	Cu175	95	8	64.40420	25.65445	27.37052
Evit200	Cu000	95	8	58.30735	24.46017	24.66947
Evit200	Cu035	108	9	58.63146	23.79650	25.04444
Evit200	Cu175	96	8	60.97602	25.07763	25.93749

- Mean weights are similar across all treatment groups
- Standard deviations are also similar
 - Variability is consistent across groups
- Similar observations and sample sizes across groups
- Cu175 seems to have slightly elevated mean weight when Evit is present

Model Output

```
Linear mixed model fit by REML. t-tests use Satterthwaite's method ['lmerModLmerTest']  
Formula: Weight ~ Evit * Cu * Time + (1 | Litter/Pig)  
Data: dietox
```

```
REML criterion at convergence: 4353.2
```

```
Scaled residuals:
```

Min	1Q	Median	3Q	Max
-4.5095	-0.4908	-0.0259	0.4855	3.7445

```
Random effects:
```

Groups	Name	Variance	Std.Dev.
Pig:Litter	(Intercept)	34.290	5.856
Litter	(Intercept)	7.713	2.777
Residual		10.184	3.191

```
Number of obs: 861, groups: Pig:Litter, 72; Litter, 21
```

- Most variation is at pig level
- Litter level variation is small
- Some variation within pigs over time (residual)

Percentage of Variance

Pig:Litter - 65.7%

Litter - 14.8%

Residual - 19.5%

Fixed Effects

Fixed effects:					
	Estimate	Std. Error	df	t value	Pr(> t)
(Intercept)	25.80820	2.67964	93.90911	9.631	1.13e-15 ***
EvitEvit100	-0.58674	3.58669	85.15183	-0.164	0.8704
EvitEvit200	-0.92540	3.56267	82.85914	-0.260	0.7957
CuCu035	-1.33695	3.58749	85.33332	-0.373	0.7103
CuCu175	-0.25326	3.58828	85.54825	-0.071	0.9439
Time2	4.05714	1.70575	690.00528	2.379	0.0177 *
Time3	10.24285	1.70575	690.00528	6.005	3.10e-09 ***
Time4	16.99999	1.70575	690.00528	9.966	< 2e-16 ***
Time5	24.51428	1.70575	690.00528	14.372	< 2e-16 ***
Time6	31.19999	1.70575	690.00528	18.291	< 2e-16 ***
Time7	37.85713	1.70575	690.00528	22.194	< 2e-16 ***
Time8	45.79999	1.70575	690.00528	26.850	< 2e-16 ***
Time9	53.05711	1.70575	690.00528	31.105	< 2e-16 ***
Time10	60.22856	1.70575	690.00528	35.309	< 2e-16 ***
Time11	69.47141	1.70575	690.00528	40.728	< 2e-16 ***
Time12	74.85712	1.70575	690.00528	43.885	< 2e-16 ***
EvitEvit100:CuCu035	2.81829	4.95761	83.03465	0.568	0.5712
EvitEvit200:CuCu035	2.18316	4.91147	84.99620	0.445	0.6578
EvitEvit100:CuCu175	2.03513	4.98963	85.38297	0.408	0.6844
EvitEvit200:CuCu175	0.79751	4.95287	82.46832	0.161	0.8725
EvitEvit100:Time2	0.36786	2.33569	690.00528	0.157	0.8749
EvitEvit200:Time2	-0.13214	2.33569	690.00528	-0.057	0.9549
EvitEvit100:Time3	-0.15536	2.33569	690.00528	-0.067	0.9470
EvitEvit200:Time3	0.19465	2.33569	690.00528	0.083	0.9336
EvitEvit100:Time4	0.93751	2.33569	690.00528	0.401	0.6883
EvitEvit200:Time4	-1.16250	2.33569	690.00528	-0.498	0.6188
EvitEvit100:Time5	-0.87678	2.33569	690.00528	-0.375	0.7075
EvitEvit200:Time5	-2.52678	2.33569	690.00528	-1.082	0.2797
EvitEvit100:Time6	-0.81249	2.33569	690.00528	-0.348	0.7281
EvitEvit200:Time6	-3.39999	2.33569	690.00528	-1.456	0.1459
EvitEvit100:Time7	0.73036	2.33569	690.00528	0.313	0.7546
EvitEvit200:Time7	-1.95715	2.33569	690.00528	-0.838	0.4024
EvitEvit100:Time8	0.85001	2.33569	690.00528	0.364	0.7160
EvitEvit200:Time8	-2.55000	2.33569	690.00528	-1.092	0.2753
EvitEvit100:Time9	0.53038	2.33569	690.00528	0.227	0.8204
EvitEvit200:Time9	-2.13213	2.33569	690.00528	-0.913	0.3616
EvitEvit100:Time10	1.32141	2.33569	690.00528	0.566	0.5717
EvitEvit200:Time10	-1.05357	2.33569	690.00528	-0.451	0.6521
EvitEvit100:Time11	-0.50892	2.33569	690.00528	-0.218	0.8276
EvitEvit200:Time11	-2.68392	2.33569	690.00528	-1.149	0.2509
EvitEvit100:Time12	0.43928	2.37770	690.09687	0.185	0.8535
EvitEvit200:Time12	-2.08742	2.37770	690.09686	-0.878	0.3803

CuCu035:Time2	-0.14464	2.33569	690.00528	-0.062	0.9506
CuCu175:Time2	1.05536	2.33569	690.00528	0.452	0.6515
CuCu035:Time3	-0.38036	2.33569	690.00528	-0.163	0.8707
CuCu175:Time3	0.79464	2.33569	690.00528	0.340	0.7338
CuCu035:Time4	-0.38749	2.33569	690.00528	-0.166	0.8683
CuCu175:Time4	0.41251	2.33569	690.00528	0.177	0.8599
CuCu035:Time5	-0.98929	2.33569	690.00528	-0.424	0.6720
CuCu175:Time5	0.34821	2.33569	690.00528	0.149	0.8815
CuCu035:Time6	-0.92500	2.33569	690.00528	-0.396	0.6922
CuCu175:Time6	0.96250	2.33569	690.00528	0.412	0.6804
CuCu035:Time7	0.16786	2.33569	690.00528	0.072	0.9427
CuCu175:Time7	2.24284	2.33569	690.00528	0.960	0.3373
CuCu035:Time8	-0.30000	2.33569	690.00528	-0.128	0.8978
CuCu175:Time8	0.81250	2.33569	690.00528	0.348	0.7280
CuCu035:Time9	0.31788	2.33569	690.00528	0.136	0.8918
CuCu175:Time9	0.36787	2.33569	690.00528	0.158	0.8749
CuCu035:Time10	-0.71608	2.33569	690.00528	-0.307	0.7593
CuCu175:Time10	0.52143	2.33569	690.00528	0.223	0.8234
CuCu035:Time11	-2.65894	2.33569	690.00528	-1.138	0.2554
CuCu175:Time11	-1.98392	2.33569	690.00528	-0.849	0.3960
CuCu035:Time12	-0.54465	2.33569	690.00528	-0.233	0.8157
CuCu175:Time12	-0.81964	2.33569	690.00528	-0.351	0.7258
EvitEvit100:CuCu035:Time2	-0.74286	3.24765	690.00528	-0.229	0.8191
EvitEvit200:CuCu035:Time2	0.25297	3.20380	690.00528	0.079	0.9371
EvitEvit100:CuCu175:Time2	-0.39286	3.24765	690.00528	-0.121	0.9038
EvitEvit200:CuCu175:Time2	-1.14286	3.24765	690.00528	-0.352	0.7250
EvitEvit100:CuCu035:Time3	0.56965	3.24765	690.00528	-0.175	0.8608
EvitEvit200:CuCu035:Time3	-0.39047	3.20380	690.00528	-0.122	0.9030
EvitEvit100:CuCu175:Time3	1.31786	3.24765	690.00528	0.406	0.6850
EvitEvit200:CuCu175:Time3	-0.00714	3.24765	690.00528	-0.002	0.9982
EvitEvit100:CuCu035:Time4	-2.66251	3.24765	690.00528	-0.820	0.4126
EvitEvit200:CuCu035:Time4	-0.29445	3.20380	690.00528	-0.092	0.9268
EvitEvit100:CuCu175:Time4	0.38749	3.24765	690.00528	0.119	0.9051
EvitEvit200:CuCu175:Time4	1.17500	3.24765	690.00528	0.362	0.7176
EvitEvit100:CuCu035:Time5	-1.07322	3.24765	690.00528	-0.330	0.7412
EvitEvit200:CuCu035:Time5	0.95734	3.20380	690.00528	0.299	0.7652
EvitEvit100:CuCu175:Time5	2.06429	3.24765	690.00528	0.636	0.5252
EvitEvit200:CuCu175:Time5	1.20178	3.24765	690.00528	0.370	0.7115
EvitEvit100:CuCu035:Time6	-1.65001	3.24765	690.00528	-0.508	0.6116
EvitEvit200:CuCu035:Time6	0.96944	3.20380	690.00528	0.303	0.7623
EvitEvit100:CuCu175:Time6	1.20001	3.24765	690.00528	0.370	0.7119

EvitEvit200:CuCu175:Time6	1.52500	3.24765	690.00528	0.470	0.6388
EvitEvit100:CuCu035:Time7	-2.99287	3.24765	690.00528	-0.922	0.3571
EvitEvit200:CuCu035:Time7	-0.17897	3.20380	690.00528	-0.056	0.9555
EvitEvit100:CuCu175:Time7	-0.41784	3.24765	690.00528	-0.129	0.8977
EvitEvit200:CuCu175:Time7	-0.83034	3.24765	690.00528	-0.256	0.7983
EvitEvit100:CuCu035:Time8	-2.16251	3.24765	690.00528	-0.666	0.5057
EvitEvit200:CuCu035:Time8	1.31666	3.20380	690.00528	0.411	0.6812
EvitEvit100:CuCu175:Time8	-0.53750	3.24765	690.00528	-0.166	0.8686
EvitEvit200:CuCu175:Time8	0.13749	3.24765	690.00528	0.042	0.9662
EvitEvit100:CuCu035:Time9	-2.81787	3.24765	690.00528	-0.868	0.3859
EvitEvit200:CuCu035:Time9	-0.48731	3.20380	690.00528	-0.152	0.8791
EvitEvit100:CuCu175:Time9	1.11964	3.24765	690.00528	0.345	0.7304
EvitEvit200:CuCu175:Time9	0.28213	3.24765	690.00528	0.087	0.9308
EvitEvit100:CuCu035:Time10	-4.05891	3.24765	690.00528	-1.250	0.2118
EvitEvit200:CuCu035:Time10	-0.23672	3.20380	690.00528	-0.074	0.9411
EvitEvit100:CuCu175:Time10	0.69110	3.24765	690.00528	0.213	0.8315
EvitEvit200:CuCu175:Time10	0.67856	3.24765	690.00528	0.209	0.8346
EvitEvit100:CuCu035:Time11	1.62142	3.24765	690.00528	0.499	0.6178
EvitEvit200:CuCu035:Time11	0.61587	3.20380	690.00528	0.192	0.8476
EvitEvit100:CuCu175:Time11	5.12142	3.24765	690.00528	1.577	0.1153
EvitEvit200:CuCu175:Time11	3.18392	3.24765	690.00528	0.980	0.3272
EvitEvit100:CuCu035:Time12	0.41072	3.27799	690.05348	0.125	0.9003
EvitEvit200:CuCu035:Time12	-1.71397	3.23455	690.05478	-0.530	0.5964
EvitEvit100:CuCu175:Time12	2.56153	3.30804	690.11326	0.774	0.4390
EvitEvit200:CuCu175:Time12	0.74992	3.27799	690.05347	0.229	0.8191

● Only Time fixed effects are statistically significant with p-values close to 0

ANOVA Results

Effect	Sum Sq	Mean Sq	NumDF	DenDF	F value	Pr(>F)
Evit	2.037138e+01	10.185692	2	54.80631	1.0002137	0.3744144
Cu	1.874100e+01	9.370499	2	54.14823	0.9201635	0.4045935
Time	4.914531e+05	44677.550349	11	690.01299	4387.2423323	0.0000000
Evit:Cu	7.415566e+00	1.853891	4	55.19439	0.1820483	0.9467644
Evit:Time	2.483229e+02	11.287404	22	690.01597	1.1083995	0.3312380
Cu:Time	1.440213e+02	6.546424	22	690.01571	0.6428452	0.8941707
Evit:Cu:Time	1.617607e+02	3.676381	44	690.01635	0.3610129	0.9999608

- Time is very significant, $p = 0$
- Evit and Cu are not significant, $p > 0.35$
- Interactions also not significant

Estimated Marginal Means - Time

Time	emmean	SE	df	lower.CL	upper.CL
1	25.66805	0.8356028	104.5944	24.01113	27.32497
2	29.88194	0.8356028	104.5944	28.22502	31.53886
3	36.09305	0.8356028	104.5944	34.43612	37.74997
4	42.42082	0.8356028	104.5944	40.76390	44.07774
5	49.14860	0.8356028	104.5944	47.49168	50.80552
6	55.65693	0.8356028	104.5944	54.00001	57.31385
7	63.40137	0.8356028	104.5944	61.74445	65.05829
8	70.91248	0.8356028	104.5944	69.25556	72.56940
9	78.17637	0.8356028	104.5944	76.51945	79.83329
10	85.56803	0.8356028	104.5944	83.91111	87.22495
11	93.63331	0.8356028	104.5944	91.97639	95.29023
12	99.66487	0.8394337	106.4734	98.00069	101.32904

- Weight increases steadily week by week
- Growth is approximately linear
 - Around 6kg to 8kg per week
- No overlap in confidence intervals
- Supports the strong time effect seen in ANOVA results

Estimated Marginal Means - Treatment

Evit	Cu	emmean	SE	df	lower.CL	upper.CL
Evit000	Cu000	61.49867	2.456245	62.54921	56.58956	66.40777
Evit100	Cu000	61.14720	2.300315	62.86319	56.55019	65.74421
Evit200	Cu000	58.94902	2.301172	62.88773	54.35034	63.54771
Evit000	Cu035	59.61499	2.298601	62.77690	55.02128	64.20870
Evit100	Cu035	60.69029	2.302444	62.90400	56.08908	65.29149
Evit200	Cu035	59.31604	2.169475	62.95168	54.98062	63.65145
Evit000	Cu175	61.63827	2.301305	62.87405	57.03930	66.23724
Evit100	Cu175	64.41485	2.300957	62.87756	59.81658	69.01313
Evit200	Cu175	60.46559	2.300194	62.84554	55.86880	65.06238

- Means across groups controlling for time
- Means are very close across all groups, between 58kg and 64kg
- Heavy overlap in confidence intervals
- Cu175 still looks slightly higher but this is *not* statistically significant
- No meaningful difference between treatment groups

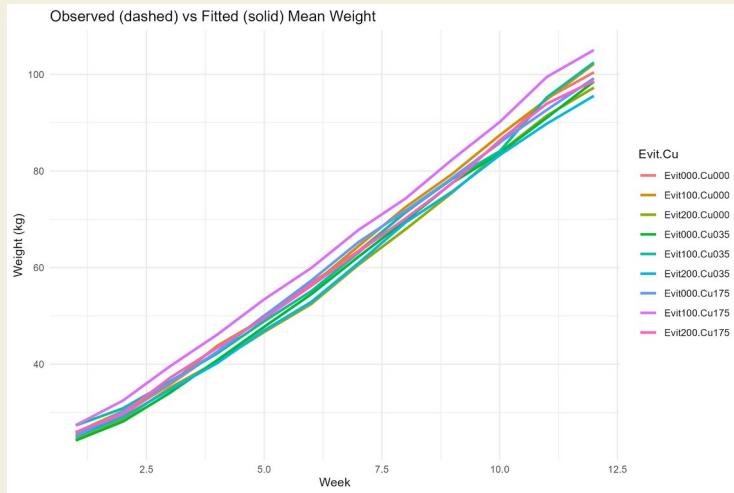
Tukey Adjusted Pairwise Comparisons

contrast	estimate	SE	df	t.ratio	p.value
Evit000 Cu000 - Evit100 Cu000	0.3514700	3.278772	55.44799	0.10719563	1.0000000
Evit000 Cu000 - Evit200 Cu000	2.5496423	3.241682	53.58208	0.78651836	0.9968086
Evit000 Cu000 - Evit000 Cu035	1.8836768	3.279470	55.56835	0.57438452	0.9996681
Evit000 Cu000 - Evit100 Cu035	0.8083815	3.282670	56.04615	0.24625733	0.9999995
Evit000 Cu000 - Evit200 Cu035	2.1826303	3.159847	53.55375	0.69073931	0.9987202
Evit000 Cu000 - Evit000 Cu175	-0.1396011	3.280028	55.72232	-0.04256096	1.0000000
Evit000 Cu000 - Evit100 Cu175	-2.9161882	3.279243	55.50779	-0.88928691	0.9926932
Evit000 Cu000 - Evit200 Cu175	1.0330779	3.279564	55.67945	0.31500469	0.9999967
Evit100 Cu000 - Evit200 Cu000	2.1981723	3.134628	53.39774	0.70125450	0.9985738
Evit100 Cu000 - Evit000 Cu035	1.5322069	3.172249	55.74311	0.48300332	0.9999095
Evit100 Cu000 - Evit100 Cu035	0.4569115	3.175958	56.33421	0.14386572	1.0000000
Evit100 Cu000 - Evit200 Cu035	1.8311603	3.056208	54.41168	0.59916091	0.9995452
Evit100 Cu000 - Evit000 Cu175	-0.4910711	3.177515	56.48441	-0.15454567	1.0000000
Evit100 Cu000 - Evit100 Cu175	-3.2676582	3.138288	53.87315	-1.04122300	0.9798039
Evit100 Cu000 - Evit200 Cu175	0.6816080	3.180220	56.86130	0.21432733	0.9999998
Evit200 Cu000 - Evit000 Cu035	-0.6659655	3.173640	55.93502	-0.20984281	0.9999999
Evit200 Cu000 - Evit100 Cu035	-1.7412608	3.213737	58.33916	-0.54181814	0.9997867

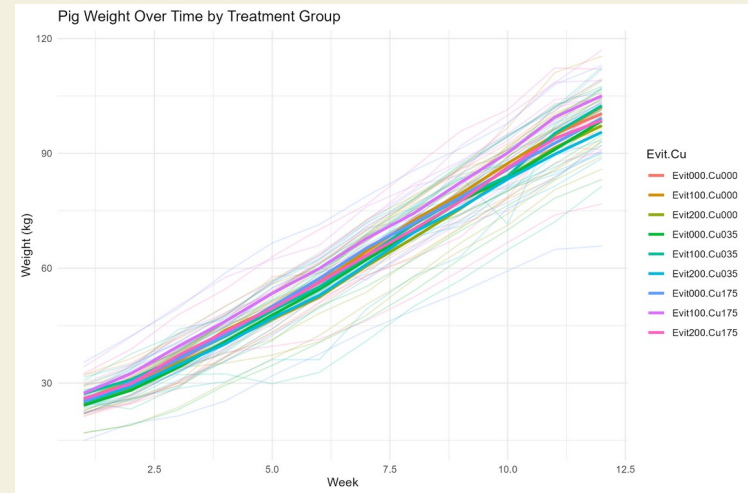
Evit200 Cu000 - Evit200 Cu035	-0.3670120	3.059629	54.86098	-0.11995310	1.0000000
Evit200 Cu000 - Evit000 Cu175	-2.6892434	3.179876	56.80275	-0.84570699	0.9947977
Evit200 Cu000 - Evit100 Cu175	-5.4658305	3.138781	53.94707	-1.74138617	0.7187145
Evit200 Cu000 - Evit200 Cu175	-1.5165644	3.173857	56.10659	-0.47783003	0.9999167
Evit000 Cu035 - Evit100 Cu035	-1.0752953	3.101447	51.64748	-0.34670761	0.9999928
Evit000 Cu035 - Evit200 Cu035	0.2989534	3.086924	56.04985	0.09684509	1.0000000
Evit000 Cu035 - Evit000 Cu175	-2.0232780	3.100244	51.47904	-0.65261899	0.9991439
Evit000 Cu035 - Evit100 Cu175	-4.7998650	3.139530	54.10730	-1.52884806	0.8369652
Evit000 Cu035 - Evit200 Cu175	-0.8505989	3.139804	54.22605	-0.27090824	0.9999990
Evit100 Cu035 - Evit200 Cu035	1.3742488	3.093123	56.90976	0.44429165	0.9999522
Evit100 Cu035 - Evit000 Cu175	-0.9479826	3.029673	47.10891	-0.31289928	0.9999967
Evit100 Cu035 - Evit100 Cu175	-3.7245697	3.177321	56.46983	-1.17223580	0.9591021
Evit100 Cu035 - Evit200 Cu175	0.2246964	3.138001	54.02471	0.07160496	1.0000000
Evit200 Cu035 - Evit000 Cu175	-2.3222314	3.061660	55.26172	-0.75848779	0.9975310
Evit200 Cu035 - Evit100 Cu175	-5.0988184	3.057113	54.51574	-1.66785432	0.7627371
Evit200 Cu035 - Evit200 Cu175	-1.1495523	3.088094	56.13053	-0.37225302	0.9999877
Evit000 Cu175 - Evit100 Cu175	-2.7765870	3.210761	57.96830	-0.86477546	0.9939690
Evit000 Cu175 - Evit200 Cu175	1.1726791	3.136347	53.77246	0.37389963	0.9999872
Evit100 Cu175 - Evit200 Cu175	3.9492661	3.177058	56.47897	1.24305779	0.9429909

● No pairwise comparisons are statistically significant

Plots cont.



- Fitted mean weight very closely follows the observed values
- Close fit is driven by strong time effect



- Pig weight over time very similar across all treatment groups, suggesting that time is most important

Conclusion

Pig weight increases significantly over time (RQ1)

No statistically significant effect of Vitamin E or Copper on weight (RQ2)

No statistically significant interaction between Vitamin E or Copper (RQ3)

Estimated marginal means show similar average weights/confidence intervals across groups (RQ4)

Residuals show non-normality in tails despite transformations, however homogeneity of variances passes

Existing Results

- The original paper (Lauridsen et al.) also found no significant effects on weight from copper or vitamin E
- However, this feed did have effects on the oxidative behaviors of the pigs as well as its liver composition
- The weight result indicates that adding copper and vitamin E to the food can have positive effects on the health of pigs without affecting how the pig grows

Questions?